dfnWorks → GEOS Workbench

Windows quick tutorial — input in, results out

The workbench turns a dfnWorks fracture network into a **GEOS-ready model**: a VTK mesh with an explicit fracture block, and the XML input deck(s) that run it. You feed it files through a small web panel and download the results.

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1. Install once

1. **Python 3.10+** — https://python.org (tick *Add python.exe to PATH*).
2. **Docker Desktop** — https://docker.com/products/docker-desktop, keep the **WSL2 backend** enabled.
3. Open *PowerShell* and install the Python dependency, then the dfnWorks image:

PowerShell / cmd

```
pip install numpy
docker pull ees16/dfnworks:latest
```

> Docker must be running (whale icon in the tray) before you start a job.

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2. Start the panel

PowerShell / cmd

```
cd C:\path\to\dfn-geos-workbench
python run_panel.py
```

Open **http://localhost:8765**. Keep PowerShell open; `Ctrl+C` stops the panel. Cases, logs and the database are written next to the code (`cases\` , `panel.db`).

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3. Configure it once

Click **Settings** and fill in:

Field	Value on Windows
dfnWorks backend	`docker`
dfnWorks docker image	`ees16/dfnworks:latest`
dfnWorks repository	`C:\path\to\dfnWorks` (only needed for native runs)
GEOS executable	leave empty (see step 6)

Save. The dashboard now shows the resolved paths.

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4. Give it your input

Click **Quick process** in the top bar and either

- **drop a ZIP** on the page, or

- **pick loose files** (they are packed automatically).

Accepted content — the panel detects it and runs the right stage by itself:

Your files	What it does
`full_mesh.inp`, `params.txt`, `polys\`, `dfm_mesh\`	converts to a GEOS case
`dfn_driver.py`, `input_generator.dat`, `case_params.json`	generates the dfnWorks mesh, then converts
`*.xml` + `*.vtm` / `*.vtu`	runs GEOS on your own deck
nothing recognised	runs the full pipeline with defaults

Give the case a name (optional) → **Process**. Stages chain automatically and the page refreshes while they run.

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5. Collect the output

Open the case and click ⬇ **Download results (.zip)**. You get:

```
geos_case\
  case.vtm                  multi-block container GEOS reads
  main.vtu                  rock matrix (tetrahedra, region attribute, GlobalIds)
  fracture.vtu              explicit fracture surfaces (triangles, aperture,
                            fracture_id, collocated_nodes)
  geos_dfn_fractures.xml    conformal GEOS deck
geos_case_edfm\
  geos_dfn_edfm.xml         EDFM GEOS deck (the runnable one today)
output\                     dfnWorks files produced or uploaded
```

The **Artifacts** table lists each generated file; **Files** browses everything; `log` next to any job shows the full stdout for diagnosis.

> `geos_case_edfm\geos_dfn_edfm.xml` is the deck to use: it represents the whole fracture network as embedded surface elements inside the rock matrix.

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6. Optional — also run GEOS

Stage *3 · Run GEOS* needs a GEOS binary (GEOS ships no Windows build):

- **WSL2** — build once inside Ubuntu-on-WSL, then point *GEOS executable* at the Linux path and set *GEOS backend* to `native`;
- **Docker** — build a GEOS image once and set *GEOS backend* = `docker`, *GEOS image* = your tag, *GEOS executable* = `/opt/geos/bin/geosx`.

Step-by-step commands are in `docs\INSTALL_windows.md` §5. Without it, everything up to the decks still works.

When it runs, GEOS writes `*.pvd` + per-step `*.vtu` files; they appear under **Artifacts** and in the results ZIP.

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7. If something goes wrong

Symptom	Fix
`docker: command not found`	Start Docker Desktop and reopen PowerShell
Stage 1: *no intersecting fractures*	Enlarge fractures or lower *Mesh h* in the case form
Stage 1: `MO DOES NOT EXIST: mo_poisson_pts`	Using a released LaGriT; switch backend to `docker`
Stage 3: *GEOS executable is not configured*	Expected without step 6
Output files owned by `root`	Run the case from Docker Desktop's shared drive, or `docker run --rm -v "%cd%":/w ees16/dfnworks:latest chown -R 1000:1000 /w`
Panel port busy	`python run_panel.py --port 9000`

Command-line equivalents and the full reference: `docs\USER_GUIDE.md`, `docs\INSTALL_windows.md`, `report\feasibility_report.md`.